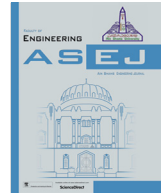




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The role of implementing BIM applications in enhancing project management knowledge areas in Egypt

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ABSTRACT

Egypt's construction industry suffers from poor management, which causes waste of time, cost, and material. Many techniques and tools have been developed to improve the project management process and deal with challenges to meet stakeholders' expectations. In Egypt, BIM technology would be efficient in managing projects at any stage during the project lifecycle if the construction processes have been linked to BIM concepts. The aims of this study are: to explore the actual status of applying project management knowledge areas in the construction industry sector in Egypt, to investigate the real impact of using BIM technology as a tool in enhancing the application of project management knowledge areas, and to explore the real benefits have been achieved from implementing BIM applications in project management. To achieve the research objectives, the researcher used the descriptive, analytical, and quantitative approach to reach conclusions. This study was applied in Egypt and was limited to those who have experience in using BIM applications in project management, whether preparing drawings, details, estimating quantities, cost management, managing, extracting the schedules etc. The research sample consisted of (106) respondents. It was concluded that Implementing BIM applications in project management could ultimately enhance the project management process and encompass all areas of management unevenly. Many benefits are defined, such as Centralized data management and information flow, improving cost control, maintaining the construction schedule, descriptive information (Object, and it is information), and Dealing with stakeholders with different backgrounds.

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1. Introduction

Egypt's construction industry suffers from poor management, which causes waste of time, cost, and material [1,2]. Project management tools and methods have not been implemented effectively in Egypt's construction projects by contracting companies [3]. Nowadays, project management is considered a competitive tool to achieve the required quality [4]. Managing projects cover many aspects during the project lifecycle in order to increase efficiency. Many techniques and tools have been developed to improve the project management process and deal with challenges

to meet stakeholders' expectations [5]. Managing a construction project needs various skills, applications, and techniques.

However, the construction sector in Egypt is backward in implementing new technologies and does not encourage utilizing BIM technologies that improve project management quality. Moreover, there is no explicit knowledge of BIM applications [6]. In Egypt, BIM technology would be efficient in managing projects at any stage during the project lifecycle if the construction processes have been linked to BIM concepts [7].

Many industries have adopted an intelligent industry trend that aims to integrate the physical world with the digital one to enhance the intelligent industry's productivity. It provides big data analytics and smart data acquisition [8]. BIM is defined as an integrated representation of a building, process-related information, and encapsulating geometry that enables project stakeholders to work collaboratively throughout the project's lifecycle [9]. The BIM handbook defines BIM as a phrase to describe tools, processes, and technologies facilitated by digital machine-readable documentation about building performance, planning, construction, and operation [9]. BIM is a digitalized technology used to create,

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arrange, oversee, trade and share building information in an inter-operable way [10]. BIM is also defined as “a set of interacting policies, processes and technologies generating a methodology to manage the essential building design and project data in digital format throughout the building’s lifecycle” [11].

BIM technology nowadays is used in visualization, design, project construction planning, scheduling, and cost estimating purposes. Many BIM software programs were developed as modelling tools such as Revit and ArchiCAD, which transfer construction project drawings and documents from traditional case to object-oriented 3D models based on information that describe building in details [12]. The prospect of using this technology goes beyond all the bounds of data gathering and analysis but managing it between all stakeholders [13].

The sense of the research problem stemmed from several matters, namely:

1. Through extrapolation of previous studies which discussed the benefits of implementing BIM innovations in the construction industry and recommend BIM as an effective management tool that deals with a significant amount of information.
2. The lack of clear information related to the BIM adoption situation and processes that use this technology in managing construction projects in Egypt and implementing it.
3. Lack of studies that promote the use of BIM in the construction industry in Egypt.

However, studies related to the impact of using BIM in managing construction projects in Egypt are limited. That’s why the main aims of this study are as mentioned below:

1. The first aim is to explore the actual status of applying project management knowledge areas in the construction industry sector in Egypt
2. The second aim is to investigate the real impact of using BIM technology to enhance the application of project management knowledge areas depending on participant’s experiences.
3. The study also explores the real benefits of implementing BIM applications in project management from the participant’s point of view then evaluating and ranking those benefits according to their importance depending on the participant’s experiences.
4. This research aims finally to provide a future vision depending on the current situation for the development of project management methods and tools by integrating them with innovations to ensure the least waste of time, cost, and effort and to increase quality and to improve productivity.

Data was collected from one hundred and six project managers who have experience using BIM applications in managing projects and so on engineers involved in designing, constructing, and

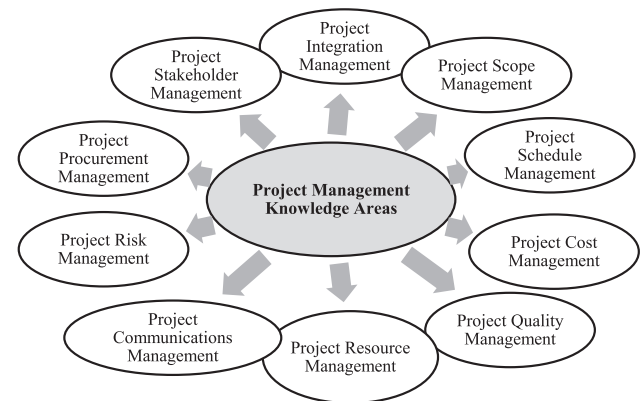


Fig. 1. Project Management Knowledge Areas [15].

managing projects using BIM applications at different project stages. The research aims were achieved through literature review and collecting qualitative data from respondents.

2. Literature review

2.1. Project management knowledge areas

PMBok distributes ten project management knowledge areas: project integration, scope, cost, time, quality, human resource, communications, procurement, risk, and stakeholder management [14]. Fig. 1 presents the project management knowledge areas.

Each project management Knowledge Area broken into three main tasks depending on PMBOK® Guide such as “project integration management”, which consists of plan development, execution, and overall change control [14]. The complete managing tasks are thirty after merging by the researcher, as shown in Table 1.

2.2. Background of BIM

As new technology in designing and construction projects, BIM can provide an intelligent 3D object-oriented database representing integrated information and a relational database that is interconnected [4]. BIM is known as n-D modeling application which include 3D modelling, project schedule, costing tool, and management aided tool that affects communication effectively and enhancing collaboration between project stakeholders [16]. BIM as an innovative technology offers a new vision to the architecture, engineering and construction industry (AEC)⁽¹⁾ and has many benefits [17]. On the other side, stakeholders face many challenges when implementing this facility, such as lack of national standards, training engineers, copyright protection for ownership of data, software-related issues, and computer systems’ ability to exchange

Table 1
Project Management Knowledge Areas and their main tasks [15].

Integration Mgmt.	Scope Mgmt.	Schedule Mgmt.	Cost Mgmt.	Quality Mgmt.
1. Develop Project Management Plan 2. Perform Integrated Change Control 3. Monitor and Control Project Work	1. Plan and Define Scope 2. Validate Scope 3. Control Scope	1. Plan Schedule (Define, Sequence, and Estimate Durations for Activities) 2. Develop Schedule 3. Control Schedule	1. Plan Cost Management 2. Estimate Costs 3. Determine and Control Budget Procurement Mgmt.	1. Plan Quality Management 2. Manage Quality 3. Control Quality
Resource Mgmt.	Communications Mgmt.	Risk Mgmt.		Stakeholder Mgmt.
1. Plan Resource 2. Acquire, Develop, and Manage 3. Monitor	1. Plan 2. Manage 3. Monitor	1. Plan and Identify Risk 2. Risk Analysis (Qualitative/Quantitative) 3. Risk Responses and Monitor	1. Plan 2. Conduct 3. Control	1. Identify and Plan Stakeholder Engagement 2. Manage Stakeholder Engagement 3. Monitor Stakeholder Engagement

information [18]. Many previous studies explored the critical success factors of implementing BIM in the construction industry, such as conflict detection, four dimensions construction process simulation, earlier 3D visualization of the project, and improved performance and quality of the construction project [19]. BIM considers one positive effect of information and communication technology on the construction industry in design and management [20]. Recently, BIM technologies deal with other applications like multidimensional CAD information systematically; moreover, they cover construction project stages from design to operation phases [21].

2.3. Implementing BIM in project management

In the BIM-based approach, a 3D model is prepared from plans in Revit software in order to estimate the cost of building structure elements; the quantities are generated in the form of tables [22]. BIM enhances the building industry's quality and productivity and is defined as a modelling technology that produces, analyzes, and communicates building models through the projects life cycle. The advantages of using this technology are improving project cost management [23]. The main target for project stakeholders is implementing technology that reduces project cost and time, increases quality and productivity, and reduces waste, such as BIM innovations [24,25]. BIM enables many better aspects such as better project lifecycle management, sustainable design, alternative solutions, visualization of the construction process, improvements in the tendering process, and safety management [26]. BIM could be implemented at all stages of the construction project, all project stakeholders can use it to manage and control project such as owner used it to understand project scope, consultants used it to analyze, design, and develop projects, and contractor used BIM to plan and manage the project [27].

BIM is considered one of the intelligent management approaches that deal with construction projects complexities and stakeholder expectations. Projects have several sub-tasks executed through systematized processes such as procurement, scheduling, mobilization etc., to prevent many risks as cost and time overruns.

Feedback and precision are critical factors to ensure management success. BIM aims to integrate project information to develop solutions depending on a process-based database platform that provides data transfer among all stakeholders [13]. See Fig. 2, which presents one of the BIM application -Revit interface.

2.4. Benefits of implementing BIM in project management

Using BIM in Project Management can improve construction planning and reduces rework and time waste. BIM enhance project information exchange which eliminates many problems related to lousy communication among stakeholders [28]. Many studies investigate the BIM benefits in construction project management. In Slovak, the BIM benefits in construction projects are cost reduction, eliminating documentation errors, increasing project documentation quality, improving cost control, improving communication between project stakeholders, reducing documents and drawings preparation, and facilitating decision-making [29]. BIM is considered a new technology for designing, constructing, maintaining and managing projects which improve project cost analysis efficiency [30]. Another study mentioned BIM benefits in project management, such as enhancing performance and productivity, reducing wastes, and reducing the project delivery time [31]. Technical reports from project stakeholders showed many BIM benefits such as simultaneous access to project data, vital information by facilitating collaboration virtual digital medium, interdisciplinary interactions, reduction risks of conflicts, encouraging communication, and auto-quantification to improve accuracy, quality communication, and multi-dimensional integration [32]. Many attempts to take advantage of BIM in combination with other applications to have a benefit in projects construction such as a study in Egypt examined the merging between BEM and BIM as an approach to enhance environmental policies which reveals several barriers to implement this approach [17].



Fig. 2. Rivet interface.

3. Material and methods

In order to achieve the objectives of the research, the researcher used the descriptive, analytical, and quantitative approach to reach conclusions, as it is the most appropriate approach type suitable for this research to describe an actual situation. Because it depends on the description of the reality or phenomenon as it exists. The study aims to describe the reality of applying project management and implementing BIM technology in managing projects in Egypt. A qualitative survey was conducted and targeted civil engineers, architects, mechanical and electrical engineers who have experience in using this technology in managing projects. As well, project managers also who using BIM application in one or all management knowledge areas.

Research methodology can be summarized as follows:

1. The study began with a problem definition then a desk study to explore the benefits of implementing BIM applications in project management from previous studies.
2. Then the study progressed into a survey (First questionnaire) to investigate the extent of applying project management knowledge areas in the construction industry in Egypt and to investigate also the impact of implementing BIM technology on enhancing project management.
3. After completing the first survey, the second questionnaire was designed to explore the benefits achieved from implementing BIM applications in project management
4. Results revealed a specific description of the actual situation.

This will help better address the study area with its precise nature and particular circumstances, describing and examining the actual case to suggest solutions.

3.1. Study limits and community

This study was applied in Egypt and was limited to those who have experience in using BIM applications in project management, whether preparing drawings, details, estimating quantities, cost management, managing, extracting the schedules etc.

Research results are associated with the extent of application project management knowledge areas and the efficiency of implementing BIM in managing construction projects in Egypt. A unique characteristic of the study sample was needed to reach objectives, like project managers with BIM experience. The research sample consisted of (106) respondents. Fig. 3 shows the distribution of the research sample according to the research variables.

3.2. Preparation of the research tool (Questionnaire)

Data collection was conducted by two questionnaires:

- The first questionnaire: consisted of four sections
 1. The first section: is related to personal and general information about respondents.
 2. The second section: is related to exploring the actual status of applying project management knowledge areas. Respondents were asked to evaluate project knowledge areas and their tasks according to their actual application in Egypt,
 3. The third section: is to investigate the real impact of using BIM technology as a tool in enhancing the application of project management knowledge areas depending on participants' experiences

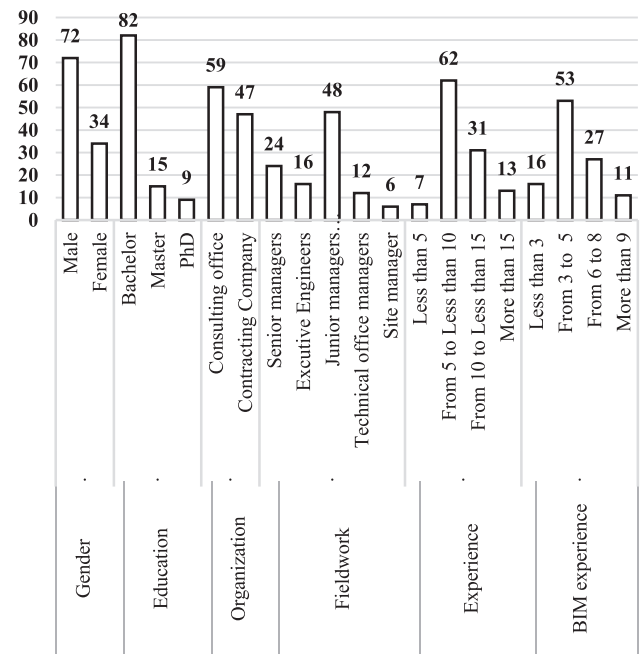


Fig. 3. Characteristics of the Study Sample.

4. The last section: is related to the benefits of implementing BIM in the construction industry in Egypt. From the respondent's point of view, they were asked to mention the services depending on their experiences. See Appendix A: The Questionnaires Forms.
- The second questionnaire: consists of one section related to explore and evaluate the benefits achieved from implementing BIM applications in project management. Benefits in this section were 64 Statements collected from the last section of the first questionnaire and the previous studies. Benefits were collected and classified, and then they were listed in a questionnaire to be evaluated to assess which of them are more effective in Egypt. Defining BIM benefits in project management will promote BIM application in managing construction projects by companies to reach the target and manage projects successfully. See Appendix A: The Questionnaires Forms.

The typical Likert five-point scale is an ordinal one used in questionnaires by respondents to rate the degree to which they agree or disagree with a statement. Although the responses can be ranked on the ordinal scale, the distance between them is not measurable. Likert scale weight is given to each statement by the respondents from 1, 2, 3, 4 and 5 for never, rarely, sometimes, often and consistently, respectively or entirely disagree, disagree, neutral, agree, and entirely agree respectively [33].

The research tool's validity was verified by presenting it to a group of specialized referees, who made some observations and modifications about the validity of the scale in studying this problem, then pre-testing it, and finally, conducting a pilot study.

After collecting the research data, the researcher converted the verbal answers into digital. SPSS statistical packages program was used to analyze data. Mean values and relative importance index were extracted. Relative Importance Index, RII, given below, is used to rank variables within groups [34]:

$$RII = \frac{\text{The summation of the total points score}}{5 \times N} \quad (1)$$

The value of RII ranges from 0 to 1, which is $0 \leq RII \leq 1$. RII is used because it fits the purpose of comparing different variables [35].

4. Results and discussion

This section of the study illustrates the findings based on quantitative and qualitative surveys. The research tool (First questionnaire) was distributed to 135 respondent, and 106 questionnaires were retrieved and completed appropriately. Thus, the recovery rate is 78.5%, which is an acceptable percentage to conduct the research. The second questionnaire was distributed to the 106 respondent who fills the first one, and they all recover it.

4.1. The real application of the project management knowledge areas

The first analysis aims to measure the actual application of the project management knowledge areas in Egypt in order to achieve the research first aim, which is "The first aim related to exploring the real status of applying project management knowledge areas. Respondents were asked to evaluate project knowledge areas and their tasks according to their real application in Egypt. The five-point Likert scale from 1 to 5 was used to receive answers. The quantifiers for the rating scale graduated from the answer (Totally not applied) to the answers (Not applied), (Don't know), (applied), to the answer (Totally applied), respectively. The question was, "what is the extent of actual application of the knowledge areas of project management in Egypt's construction sector? Give a rate from 1 to 5". Table 2 presents the analysis results, which show the ranking of the project management knowledge areas according to their actual application from the respondent's point of view using mean value and RII.

Table 2 shows that the management area, which is mainly applied and received attention from construction project managers ranked in the first position, is "Quality management" with RII equals 78%, followed by schedule, cost, and scope management. "Project integration management" comes at last position with RII equals 63%.

4.2. The impact of using BIM technology as a tool in applying project management knowledge areas

The second aim is to investigate the real impact of using BIM technology to apply project management knowledge areas depending on the participant's experiences. The five points Likert scale from 1 to 5 was used to receive answers. The quantifiers for the rating scale graduated from the solution (Totally don't enhance) to the answers (don't enhance), (Don't know), (improve), to the answer (Totally enhance), respectively. The question was, "To what extent did BIM applications enhance the application of the knowledge areas of project management in the construction sector in Egypt? Give a rate from 1 to 5". Table 3 also shows the

Table 2
Real application of PMKAs in Egypt according to survey results.

Rank	PMKAs	Mean Value	RII
1	Project Quality Management	3.916	78%
2	Project Schedule Management	3.795	76%
3	Project Cost Management	3.755	75%
4	Project Scope Management	3.749	75%
5	Project Resource Management	3.679	74%
6	Project Risk Management	3.595	72%
7	Project Stakeholder Management	3.592	72%
8	Project Procurement Management	3.586	72%
9	Project Communications Management	3.293	66%
10	Project Integration Management	3.139	63%

Table 3

The current situation of real influence of using BIM Technology as a tool on applying PMKAs.

Rank	PMKAs	Mean Value	RII
1	Project Risk Management	4.247	85%
2	Project Communications Management	4.169	83%
3	Project Cost Management	4.098	82%
4	Project Schedule Management	4.062	81%
5	Project Stakeholder Management	4.058	81%
6	Project Quality Management	3.997	80%
7	Project Scope Management	3.829	77%
8	Project Resource Management	3.819	76%
9	Project Procurement Management	3.690	74%
10	Project Integration Management	3.473	69%

impact of applying project management knowledge areas using BIM applications.

Table 3 shows higher values of RII in all management areas when using BIM applications. Risk management comes at the first position with RII equals 85%. This implies that using BIM applications to apply project management enhances all project management knowledge areas.

To assess the impact of using BIM applications on project management knowledge areas by comparing the two cases (the first case is the actual application of PMKs, and the second case is implementing BIM in project management). The arithmetic mean values and the relative importance index were calculated to rank the management areas and compare after and before using BIM). The following Fig. 4 shows the difference between the two cases.

As shown in Fig. 4, using BIM applications in managing projects enhance the management process, which is seemed clear in all management knowledge areas. Project management areas that were improved more than others are project communication, risk, and stakeholder's management. The least affected project management areas using BIM applications are procurement, resources, scope, and quality management. As indicated before, each project management area broken into three main tasks depending on PMBOK® Guide; the complete managing tasks are thirty. Respondents were asked to evaluate the success level of applying these tasks from their perspective depending on their experience in managing projects.

The following Fig. 5 and Table 4 show the differences in the RII values between applying tasks of project management areas

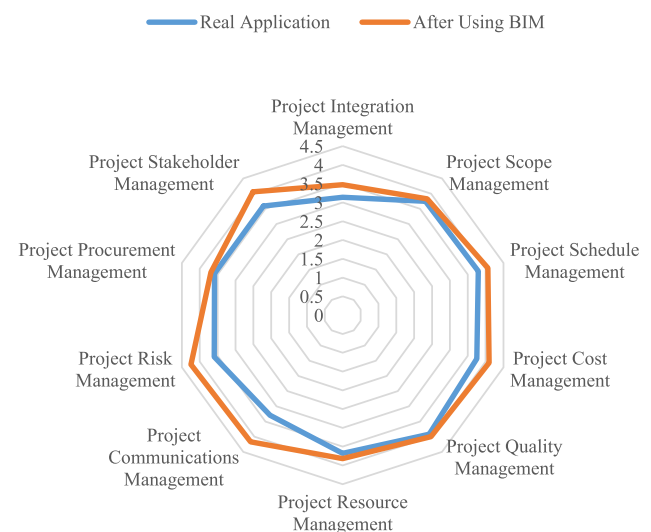


Fig. 4. The impact of using BIM technology as a tool in applying project management knowledge.

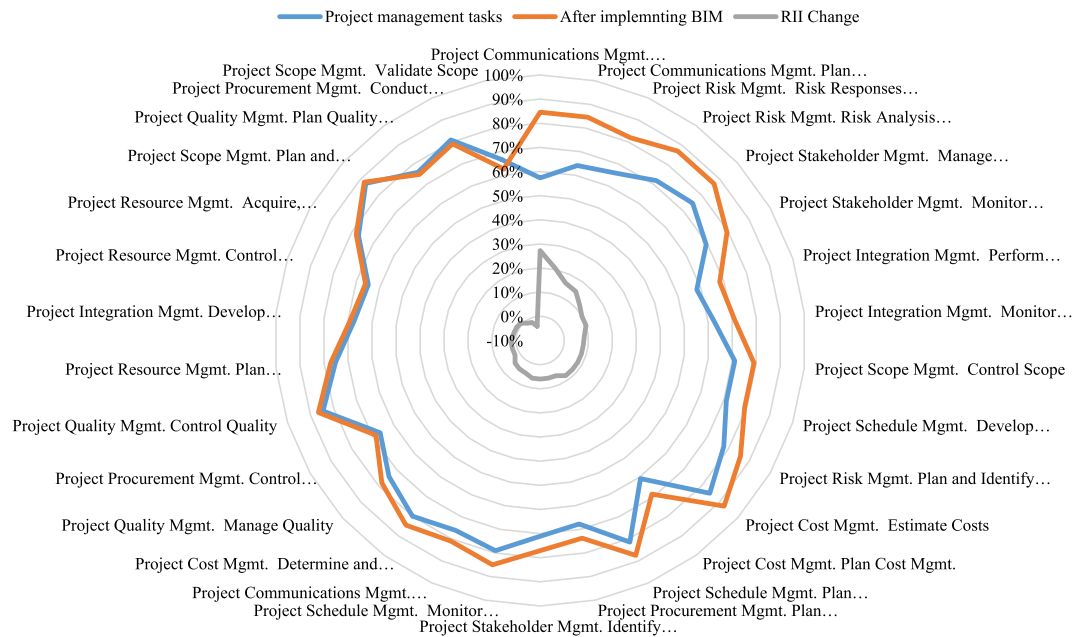


Fig. 5. The differences in the RII values between the application of project management areas tasks after and before using BIM applications.

Table 4

Ranking of BIM implementation benefits in project management knowledge areas in Egypt.

PMKAs	Variables	Application		Application using BIM		Change%	Rank
		Mean Value	RII	Mean Value	RII		
Communications	Monitor Communications	2.867	57%	4.229	85%	27%	1
Communications	Plan Communications Management	3.205	64%	4.229	85%	20%	2
Risk	Risk Responses and Monitor	3.299	66%	4.099	82%	16%	3
Risk	Risk Analysis (Qualitative and Quantitative)	3.595	72%	4.349	87%	15%	4
Stakeholder	Manage Stakeholder Engagement	3.749	75%	4.349	87%	12%	5
Stakeholder	Monitor Stakeholder Engagement	3.470	69%	3.970	79%	10%	6
Integrated	Perform Integrated Change Control	2.910	58%	3.410	68%	10%	7
Integrated	Monitor and Control Project Work	3.133	63%	3.554	71%	8%	8
Scope	Control Scope	3.554	71%	3.954	79%	8%	9
Schedule	Develop Schedule	3.554	71%	3.954	79%	8%	10
Risk	Plan and Identify Risk Management,	3.892	78%	4.292	86%	8%	11
Cost	Estimate Costs	4.229	85%	4.629	93%	8%	12
Cost	Plan Cost Management	3.036	61%	3.436	69%	8%	13
Schedule	Plan Schedule (Define, Sequence, and Estimate Activity Durations for Activities	4.072	81%	4.372	87%	6%	14
Procurement	Plan Procurement Management	3.386	68%	3.686	74%	6%	15
Stakeholder	Identify and Plan Stakeholder Engagement	3.555	71%	3.855	77%	6%	16
Resources	Monitor Resources	3.952	79%	4.252	85%	6%	17
Communications	Manage Communications	3.807	76%	4.048	81%	5%	18
Cost	Determine and Control Budget	4.000	80%	4.229	85%	5%	19
Quality	Manage Quality	3.711	74%	3.911	78%	4%	20
Procurements	Control Procurements.	3.325	67%	3.425	69%	2%	21
Quality	Control Quality	4.229	85%	4.329	87%	2%	22
Resource	Plan Resource Management	3.759	75%	3.859	77%	2%	23
Integrated	Develop Project Management Plan	3.373	67%	3.454	69%	2%	24
Resources	Control Resources	3.241	65%	3.304	66%	1%	25
Resources	Acquire, Develop, and Manage Resources	3.843	77%	3.900	78%	1%	26
Scope	Plan and Define Scope Management	4.361	87%	4.401	88%	1%	27
Quality	Plan Quality Management	3.807	76%	3.751	75%	-1%	28
Procurements	Conduct Procurements	4.048	81%	3.958	79%	-2%	29
Scope	Validate Scope	3.333	67%	3.133	63%	-4%	30

and using BIM applications in applying each task. As shown in Fig. 5, the tasks which have the most significant differences in the RII values are Monitor Communications with the difference in the RII values equals 27%, then Plan Communications Management, Risk Responses and Monitor, Risk Analysis (Qualitative and Quantitative), and Manage Stakeholder Engagement. The least differences in the RII values are Plan and Control Resources Acquire and Manage Resources, Define Scope

Management, Plan Quality Management, Conduct Procurements, and Validate Scope with differences in the RII values range from -4 to 2. The differences were negative in some values and close to the reference value, which means that there is no change or influence and since it is less than the error rate of 5% percentage. Fig. 5 shows the changing values of the RII for the thirty tasks related to the project management areas after applying BIM in management.

4.3. Significant benefits gained from implementing BIM applications in project management

The study also aims to explore the benefits achieved from implementing BIM applications in project management. The five points Likert scale from 1 to 5 was used to receive answers the quantifiers for the rating scale graduated from the answer (Totally not achieved) to the answers (Not achieved), (Don't know), (achieved), to the answer (Totally achieved) respectively. The question was, "To what extent did the mentioned benefits achieved from implementing BIM in applying knowledge areas of project management in the construction sector in Egypt? Give a rate from 1 to 5".

Sixty-four "benefits" were collected from previous studies and the first questionnaire. Table 5 shows the analysis results of the benefits of using BIM in a project according to its significance from the respondents' point of view and ranking the services depending on the arithmetic mean values and the RII.

The great benefit which can be obtained from implementing BIM in managing construction projects in Egypt is "Centralized data management and information flow" with a mean value equal 4.442 and an RII value equal 91. The second higher benefit is Enhancing cost control. The first 20 benefits have high RII values, which means that the respondents agree with the importance of these benefits from their perspective, depending on their experience in the field.

5. Conclusions

This study was conducted to investigate the extent of applying project management knowledge areas in the construction industry in Egypt and investigate the impact of implementing BIM technology on enhancing project management. The study also explored the benefits of implementing BIM applications in project management from an expert's perspective and then evaluating and ranking those benefits. It was concluded that the management knowledge areas which are more applicable and received more attention and concern in the construction industry in Egypt are quality, time, cost, and scope. Despite the focus on project management in terms of time, cost, and quality, these three areas also need to be enhanced, as all construction projects generally encounter cost and time overruns and quality problems.

According to the participants' opinions in this study, project integration and communication management frequently encountered problems and failures. From their point of view, communication management needs to be improved in terms of means and applications to ensure the project's success.

Implementing BIM applications in project management can ultimately enhance the project management process and encompass all management areas unevenly. Project risk management faces many problems in planning and identifying risk, analyzing risks (Qualitative and Quantitative), and risk responses and monitoring. The results show that risk management can be significantly improved, up to 13%, if BIM applications are used and communication management, which a large percentage can substantially enhance up to 17%.

Each management area consists of a group of tasks according to PMBOK® Guide, which were reduced to three in this study; the results show that these tasks can be enhanced in varying degrees when using BIM applications. For example, communication management consists of "Monitor Communications" that can be enhanced with a percentage up to 27%, "Plan Communications Management" that can be enhanced with a percentage up to 20%, and the last task is "Manage Communications" that can be enhanced with a percentage up to 5%.

Risk management also consists of "Risk Responses and Monitor" that can be enhanced with a percentage up to 16%, "Risk Analysis (Qualitative and Quantitative)" that can be improved with a percentage up to 15%, and the last task is "Plan and Identify Risk Management" that can be improved with a percentage up to 7%.

Most of the study participants promote the use of BIM applications in managing construction projects according to the benefits they mentioned and were re-evaluated by them. The benefits are varied and include many aspects, which include data flow, cost control, schedule, project information, stakeholders with different backgrounds, efficient use of resources, enhancing communication, improving, decision making, reduction of risks, model based quantity take-off, and maintenance process. Finally, it is the right time to take a general decision that includes all the construction sector to shift to adopting technology applications in the construction industry in all project phases, which include design, construction and operation. According to that, there is a need to enhance the performance of the construction industry by adoption these technologies.

Table 5
Benefits gained from implementing BIM in project management.

PMKAs	Expected Benefits	Mean Value	RII	Rank
Communication Mgmt.	Centralized data management and information flow	4.542	91%	1
Cost Mgmt.	Enhancing cost control	4.530	91%	2
Schedule Mgmt.	Maintain the construction schedule	4.530	91%	3
Communication Mgmt.	Descriptive information (Object and it is information)	4.530	91%	4
Stakeholders Mgmt.	Deal with stakeholders with different backgrounds	4.446	89%	5
Resource Mgmt.	Efficient use of resource and tracking	4.386	88%	6
Communication Mgmt.	Clear and concise channel (Enhancing communication)	4.349	87%	7
Cost Management	Accuracy and Realistic of cost estimation	4.277	86%	8
Communication	Decision making	4.217	84%	9
Risk Mgmt.	Reduce financial risk and risk related schedule	4.148	83%	10
Cost Mgmt.	Real monitoring for cost	4.142	83%	11
Quality	Decrease repeat errors and avoiding rework	4.133	83%	12
Scope Mgmt.	Visual scoping of work	4.108	82%	13
Risk Mgmt.	Decision making	4.096	82%	14
Risk Mgmt.	Spatial visualization	4.075	81%	15
Cost Mgmt.	Model based quantity take-off	4.048	81%	16
Schedule	Timely monitoring the progress	4.048	81%	17
Communication	Speed up understanding	4.014	80%	18
Schedule	Reduce time and mitigates delays	4.000	80%	19
Quality	Maintenance helping	3.977	80%	20

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A:. The questionnaires forms

The first questionnaire: (Implementing BIM Applications in Project Management Knowledge Areas in Egypt) <https://forms.office.com/Pages/ResponsePage.aspx?id=mZhGINWBaUeeJUFUptXR-c16yee->

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Questionnaire 1

The aims of this survey are to evaluate the real status of applying project management knowledge areas in construction projects in Egypt and to investigate the real impact of using BIM technology as a tool managing projects

Section 1: Personal Information

Gender ☐ Male ☐ Female	Academic level ☐ Bachelor ☐ Master ☐ PhD	Fieldwork working ☐ Manager ☐ Site Engineer ☐ Engineer (designer) ☐ Technical office engineer ☐ Site Engineer ☐ Site manager
Years of experience ☐ Less than 5 ☐ From 5 to Less than 10 ☐ From 10 to Less than 15 ☐ More than 15	Years of BIM experience ☐ Less than 3 ☐ From 3 to 5 ☐ From 6 to 8 ☐ More than 9	Type of organization ☐ Consulting office ☐ Contracting Company

Section 2: Project Management Knowledge Areas in the Construction projects in Egypt

Evaluate the extent to which Project Management Knowledge Areas and their tasks are applied in Egypt

1	Totally not applied	2	Not applied	3	Don't know	4	applied	5	Totally applied
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Project Management Knowledge Areas	1	2	3	4	5
Project Integration Management					
1 Develop Project Management Plan					
2 Perform Integrated Change Control					
3 Monitor and Control Project Work					
Project Scope Management					
1 Plan and Define Scope Management					
2 Validate Scope					
3 Control Scope					
Project Schedule Management					
1 Plan Schedule (Define, Sequence, and Estimate Activity Durations for Activities)					
2 Develop Schedule					
3 Control Schedule					
Project Cost Management					
1 Plan Cost Management					
2 Estimate Costs					
3 Determine and Control Budget					
Project Quality Management					
1 Plan Quality Management					
2 Manage Quality					
3 Control Quality					
Project Resource Management					
1 Plan Resource Management					
2 Acquire, Develop, and Manage Resources					
3 Monitor Resources					
Project Communications Management					
1 Plan Communications Management					
2 Manage Communications					
3 Monitor Communications					
Project Risk Management					
1 Plan and Identify Risk Management,					
2 Risk Analysis (Qualitative and Quantitative)					
3 Risk Responses and Monitor					
Project Procurement Management					
1 Plan Procurement Management					
2 Conduct Procurements					
3 Control Procurements.					
Project Stakeholder Management					
1 Identify and Plan Stakeholder Engagement					
2 Manage Stakeholder Engagement					
3 Monitor Stakeholder Engagement					

Section 3: Implementing BIM in Project Management Knowledge Areas in the Construction projects in Egypt

Evaluate the extent to which **Implementing BIM enhance** Project Management Knowledge Areas and their tasks in Egypt.

1	Totally has not been enhanced	2	Has not been enhanced	3	Don't know	4	has been enhanced	5	Totally has been enhanced
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Project Management Knowledge Areas	1	2	3	4	5
Project Integration Management					
1 Develop Project Management Plan					
2 Perform Integrated Change Control					
3 Monitor and Control Project Work					
Project Scope Management					
1 Plan and Define Scope Management					
2 Validate Scope					
3 Control Scope					
Project Schedule Management					
1 Plan Schedule (Define, Sequence, and Estimate Activity Durations for Activities)					
2 Develop Schedule					
3 Control Schedule					
Project Cost Management					
1 Plan Cost Management					
2 Estimate Costs					
3 Determine and Control Budget					
Project Quality Management					
1 Plan Quality Management					
2 Manage Quality					
3 Control Quality					
Project Resource Management					
1 Plan Resource Management					
2 Acquire, Develop, and Manage Resources					
3 Monitor Resources					
Project Communications Management					
1 Plan Communications Management					
2 Manage Communications					
3 Monitor Communications					
Project Risk Management					
1 Plan and Identify Risk Management,					
2 Risk Analysis (Qualitative and Quantitative)					
3 Risk Responses and Monitor					
Project Procurement Management					
1 Plan Procurement Management					
2 Conduct Procurements					
3 Control Procurements.					
Project Stakeholder Management					
1 Identify and Plan Stakeholder Engagement					
2 Manage Stakeholder Engagement					
3 Monitor Stakeholder Engagement					

Section 4: from your experience; mention some benefits of Implementing BIM in Project Management Knowledge Areas and their tasks in Egypt

The second questionnaire: (Benefits from implementing BIM applications in project management)

<https://forms.office.com/Pages/ResponsePage.aspx?id=mZhGINWBaUeeJfAUptXR-c16yeeDOFFq-lzXxa7bCxUNDFMkpTR1FNWFozMFdDQjI2MFpUMjNQUi4u>

Questionnaire #2:

Benefits achieved from implementing BIM applications in project management

To what extent has the mentioned benefits been achieved from implementing BIM in applying knowledge areas of project management in construction sector in Egypt.

1	Totally not achieved	2	Not achieved	3	Don't know	4	Achieved	5	Totally achieved
Benefits from implementing BIM applications in project management.									
1	Cost Management	Model based quantity take-off							
2	Cost Management	Cost planning							
3	Cost Management	Accuracy and Realistic of cost estimation							
4	Cost Management	Flexibility and Faster of cost							
5	Cost Management	Collaboration and Consistencies							
6	Cost Management	Early cost decisions							
7	Cost Management	Higher productivity							
8	Cost Management	Real monitoring for cost							
9	Cost Management	Better cost control							
10	Risk Management	Risk assessment and mitigation							
11	Risk Management	Model – based Analysis of possible hazards and Particular Risk scenarios"							
12	Risk Management	Spatial visualization							
13	Risk Management	Reduce financial risk and risk related schedule							
14	Risk Management	Reduce contingencies							
15	Risk Management	Decision making							
16	Schedule Management	Reduce time and mitigates delays							
17	Schedule Management	Maintain the construction schedule							
18	Schedule Management	Timely monitoring the progress							
19	Resource Management	Reduce conflicts by Enhancing Interdisciplinary collaboration and coordination							
20	Resource Management	Reference for conflicts							
21	Resource Management	Efficient use of resource and tracking							
22	Resource Management	Collectively make decision							
23	Resource Management	Inventory management							
24	Procurement Management	Product description and modeling							
25	Procurement Management	Contractual management for procurement							
26	Procurement Management	Integrity and Reliability of procurement							
27	Procurement Management	Good procurement plan and Limit the requisition list and Diminishing of errors							
28	Procurement Management	Managing deliveries							
29	Scope Management	Visual scoping of work							
30	Stakeholders Management	Deal with stakeholders with different backgrounds							
31	Stakeholders Management	Enhancing stakeholder involvement, communication and Fostering relationship							
32	Stakeholders Management	Client satisfaction and Trust and confidence							
33	Stakeholders Management	Positive impact on marketing							
34	Stakeholders Management	Mitigate conflicts							
35	Stakeholders Management	Core managing							
36	Stakeholders Management	Informed decisions from stakeholder							
37	Integration Management	Integrated project delivery with increase collaboration and integration							
38	Integration Management	Unified system							
39	Integration Management	Cost and schedule integration							
40	Integration Management	Visualization and immersion experience							
41	Integration Management	Information availability							
42	Integration Management	Better coordination between documents							

Benefits achieved from implementing BIM applications in project management (continued)

Benefits from implementing BIM applications in project management.			1	2	3	4	5
1	Cost Management	Model based quantity take-off					
43	Integration Management	Logic analysis					
44	Safety Management	Automated safety code checking					
45	Safety Management	Design for safety					
46	Safety Management	Safety communication					
47	Quality Management	Defect management system and recording					
48	Quality Management	Decrease repeat errors and avoiding rework					
49	Quality Management	Enhancing quality by Quality status feedback					
50	Quality Management	Physical clash detection					
51	Quality Management	Detecting failure root-cause					
52	Quality Management	Lower waste and omissions					
53	Quality Management	Timely supervising quality					
54	Quality Management	Maintenance helping					
55	Quality Management	Comprehensive quality information repository					
56	Quality Management	Efficient information utilization					
57	Communication Management	Centralized, Structured data management and Information flow					
58	Communication Management	Descriptive information (Object and it is information)					
59	Communication Management	Planning several responsibilities					
60	Communication Management	Model based approach of working					
61	Communication Management	Unique language					
62	Communication Management	Clear and concise channel (Enhancing communication)					
63	Communication Management	Speed up understanding					
64	Communication Management	Decision making					

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